

Estimation of equivalent circuit parameters of solar cells using simplified equations

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HIGHLIGHTS

- The proposed method overcomes the drawback of existing analytical approaches.
- The calculation is simplified and uses only four *I-V* characteristic parameters.
- The accuracy of the proposed method is on par with other analytical approaches.

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ABSTRACT

A simplified equation-based method for estimating the equivalent circuit parameters of solar cells is proposed. The proposed method overcomes the drawback of existing analytical approaches that require the use of *I-V* curve slopes, thereby simplifying the calculation. Simulation results indicate that the proposed method achieves mean absolute percentage errors of less than 1% and root mean square errors on the order of 0.01 A. The estimation accuracy is comparable to that of existing analytical methods, and although its applicable range is slightly narrower, it remains applicable to over 95% of solar cells conforming to the single diode model.

1. Introduction

The equivalent circuit parameters of solar cells can be used for performance evaluation, design optimization, production process monitoring, and *I-V* curve analysis. They also contribute to connection optimization and maximum power point tracking (MPPT) of photovoltaic systems. It is of great practical importance to determine these parameters accurately and quickly.

Currently, there are mainly two categories of methods for extracting the equivalent circuit parameters. One is numerical methods [1–8], and the other is analytical solution methods [9–18]. Numerical methods, such as differential evolution algorithm [1,2], dynamic bifacial power gain [3], hippopotamus optimizer [4], deep learning approach [5], JAYA algorithm [6], Bayesian inference [7], Bishop model [8], ant lion optimizer (ALO), arithmetic optimization algorithm (AOA), grasshopper optimization algorithm (GOA), particle swarm optimization (PSO), sine cosine algorithm (SCA), salp swarm algorithm (SSA), vortex search algorithm (VSA), and so on, are supposed to be more accurate and more adaptable to different kinds of solar cells and modules [13]. However they need to determine the initial value of parameters first, and some of

them may waste more time towards the global optimum, so they are not suitable for on-line fault diagnosis [15].

In contrast, analytical solution methods can derive parameters directly from experimental data without iterative optimization, and their calculations are relatively simple, making them conducive to real-time application. A key limitation of the existing analytical solution methods is their dependence on *I-V* curve to extract two parameters, namely, the reciprocal slope of *I-V* curve at the open-circuit point R_{s0} , and the reciprocal slope at the short-circuit point R_{sh0} , in addition to the four parameters (V_{oc} , I_{sc} , V_m , and I_m). This poses a significant difficulty, rendering calculation infeasible without *I-V* curve. Table 1 lists the calculation expressions of some analytical solution methods, showing that they all require the prior determination of R_{s0} and R_{sh0} .

In this paper, the *I-V* characteristic equation of solar cells is simplified under the following inherent physical approximations of solar cell behavior: extremely small reverse saturation current, relatively large shunt resistance, and relatively small series resistance, then approximate expressions for equivalent circuit parameters are derived. By simplifying the equation before derivation, the shunt resistance R_{sh} is decoupled and calculated separately. This eliminates the need to determine *I-V*

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Table 1
Calculation expressions of analytical solution methods in the references.

References	Calculation expressions (excluding I_{ph})
Reference [10]	$R_{sh} = R_{sh0}$ $n = \frac{(V_m - I_m R_s)(V_{oc} + (I_m - I_{sc})R_{sh})}{(V_m - I_m R_s)BV_T}$ $R_s = \frac{(A+B)V_m}{(A+B)I_m} + \frac{BV_T}{(A+B)I_m}$ $I_0 = \frac{(R_{sh} + R_s)I_m - V_{oc}}{R_{sh} \exp(\frac{V_{oc}}{nV_T})}$
Reference [11]	$R_{sh} = R_{sh0} - R_s$ $n = \frac{V_m + I_m R_s - V_{oc}}{V_T \ln[\frac{I_{sc}(1 + R_s/R_{sh}) - I_{sc}/R_{sh}}{I_{sc}(1 + R_s/R_{sh}) - I_{sc}/R_{sh}}]}$ $R_s = \frac{R_{sh}(V_{oc} - (nV_T - 1)I_m - R_{sh}I_m)}{(V_m - I_m R_s) - V_{oc}}$ $I_0 = \frac{(R_{sh} + R_s)I_m - V_{oc}}{R_{sh} \exp(\frac{V_{oc}}{nV_T})}$
Reference [13]	$R_{sh} = R_{sh0}$ $n = n^* - C(V_{oc} - V^*) - D(I_{sc} - I_{sc}^*)$ $R_s = R_{s0} - \frac{I_{sc} - V_{oc}/R_{sh}}{nV_T}$ $I_0 = (I_{sc} - \frac{V_{oc} - R_s I_{sc}}{R_{sh}}) \exp(-\frac{V_{oc}}{nV_T})$
Reference [14]	$R_{sh} = R_{sh0}$ $n = \frac{V_m + I_m R_s - V_{oc}}{V_T \ln[\frac{I_{sc}(1 + R_s/R_{sh}) - I_{sc}/R_{sh}}{I_{sc}(1 + R_s/R_{sh}) - I_{sc}/R_{sh}}]}$ $R_s = R_{s0} - \frac{V_m + I_m R_s - V_{oc}}{I_m + \ln(I_{sc} - I_m) - \ln(I_{sc})} I_{sc}$ $I_0 = (\frac{nV_T}{R_{sh} - R_s}) \exp(-\frac{V_{oc}}{nV_T})$
Reference [15]	$R_{sh} = [\frac{1}{R_{sh0} - R_s} - \frac{I_0}{nV_T} \exp(\frac{I_{sc} R_s}{nV_T})]^{-1}$ $A(nV_T)^2 - B(nV_T) + C = 0$ $R_s = R_{s0} - \frac{I_{sc} - I^* - V^*/R_{sh0} + (nV_T/R_{sh0})}{nV_T}$ $I_0 = (I_{sc} - I_a - \frac{V_a}{R_{sh0}}) \exp(-\frac{V_a + I_a R_s}{nV_T})$

curve slope reciprocals R_{s0} and R_{sh0} in advance, results in greatly simplified expressions, and makes the calculation much easier to implement. Therefore, the contribution of this study lies in a simpler analytical method that eliminates the need for prior determination of R_{s0} and R_{sh0} .

To verify the performance of the proposed method, the derived expressions are then applied to actual solar cells and photovoltaic modules. Then, the proposed method is compared with some existing methods in references. Finally, the applicability of the proposed method is discussed.

2. Method to extract parameters

2.1. Simplification of I-V characteristic equation

There are several models for the equivalent circuit of solar cells, including single diode model, double diode model, and multi-diode model. The single diode model is most commonly used. The equivalent circuit of single diode model consists of a current source I_{ph} , a diode D , a series resistance R_s and a shunt resistance R_{sh} , as shown in Fig. 1. The corresponding I-V characteristic equation is represented by (1).

$$I = I_{ph} - I_0 \left[\exp\left(\frac{V + IR_s}{nV_T}\right) - 1 \right] - \frac{V + IR_s}{R_{sh}} \quad (1)$$

where: I is output current, I_{ph} is photo generated current, I_0 is reverse saturation current, V is output voltage, n is ideality factor, V_T is the thermal voltage, defined as $V_T = kT/q$, k is Boltzmann constant, T is absolute temperature, q is elementary charge, R_s is series resistance, and R_{sh} is shunt resistance.

Considering the reverse saturation current I_0 is extremely small, (1) can be simplified to (2).

$$I \approx I_{ph} - I_0 \exp\left(\frac{V + IR_s}{nV_T}\right) - \frac{V + IR_s}{R_{sh}} \quad (2)$$

At the open circuit point, $I = 0$, $V = V_{oc}$, at the short circuit point, $I = I_{sc}$, $V = 0$, and at the maximum power point, $I = I_m$, $V = V_m$,

substituting these values into (2) yields (3).

$$\begin{cases} 0 \approx I_{ph} - I_0 \exp\left(\frac{V_{oc}}{nV_T}\right) - \frac{V_{oc}}{R_{sh}} \\ I_{sc} \approx I_{ph} - I_0 \exp\left(\frac{I_{sc} R_s}{nV_T}\right) - \frac{I_{sc} R_s}{R_{sh}} \\ I_m \approx I_{ph} - I_0 \exp\left(\frac{V_m + I_m R_s}{nV_T}\right) - \frac{V_m + I_m R_s}{R_{sh}} \end{cases} \quad (3)$$

Considering that I_0 is extremely small, R_{sh} is relatively large, and R_s is relatively small, then (3) can be further simplified to (4). And (4) is referred to as the simplified I-V characteristic equation.

$$\begin{cases} 0 \approx I_{ph} - I_0 \exp\left(\frac{V_{oc}}{nV_T}\right) \\ I_{sc} \approx I_{ph} \\ I_m \approx I_{ph} - I_0 \exp\left(\frac{V_m + I_m R_s}{nV_T}\right) \end{cases} \quad (4)$$

2.2. Two calculation methods for series resistance

The expressions for I_0 and R_s as functions of the ideality factor n can be derived from (4), yielding (5) and (6), respectively.

$$I_0 \approx I_{sc} / \exp\left(\frac{V_{oc}}{nV_T}\right) \quad (5)$$

$$R_s \approx [nV_T \ln\left(\frac{I_{sc} - I_m}{I_0}\right) - V_m] / I_m \quad (6)$$

For comparison purposes, substituting (5) into (6) yields (7). And (7) is referred to as the first expression for the series resistance R_s .

$$R_s \approx [(V_{oc} - V_m) + nV_T \ln\left(\frac{I_{sc} - I_m}{I_{sc}}\right)] / I_m \quad (7)$$

In addition, a second expression for the series resistance R_s can be derived through differentiation. Specifically, at the open circuit point ($I = 0$), the voltage across the current source equals the open circuit voltage, that is, $V(I_{ph}) = V_{oc}$, as illustrated in Fig. 1. By applying Kirchhoff's voltage law near this point, the relationship between the output current I and the output voltage V satisfies (8). Differentiating (8) yields (9), which can be rearranged to obtain (10).

$$IR_s + V \approx V(I_{ph}) \approx V_{oc} \quad (8)$$

$$\frac{dI}{dV} R_s + 1 \approx 0 \quad (9)$$

$$\left. \frac{dI}{dV} \right|_{V \approx V_{oc}} \approx -\frac{1}{R_s} \quad (10)$$

According to (4), near the open circuit point, the output current I and the output voltage V satisfy (11). Taking the derivative of (12) yields (13). Substituting (10) into (12) yields (13). And (13) is referred to as the second expression for the series resistance R_s .

$$I \approx I_{ph} - I_0 \exp\left(\frac{V}{nV_T}\right) \quad (11)$$

$$\left. \frac{dI}{dV} \right|_{V \approx V_{oc}} \approx -I_0 \exp\left(\frac{V_{oc}}{nV_T}\right) \frac{1}{nV_T} \approx -\frac{I_{sc}}{nV_T} \quad (12)$$

$$R_s \approx \frac{nV_T}{I_{sc}} \quad (13)$$

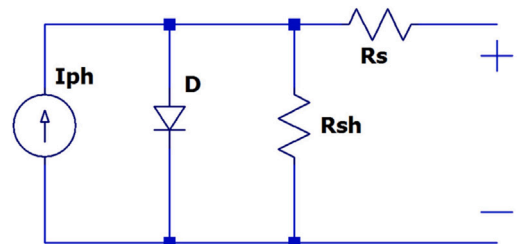


Fig. 1. Equivalent circuit diagram of solar cells.

2.3. Solution of the simplified equation

The calculation expression for the ideal factor n can be obtained, given as (14), by setting the two expressions for R_s equal to each other, i.e., (7) = (13). Once the ideality factor n is determined, the values of I_0 and R_s can be calculated using (5) and (13), respectively.

$$n \approx \frac{I_{sc}(V_{oc} - V_m)}{[I_m + I_{sc} \ln(\frac{I_{sc}}{I_{sc} - I_m})]V_T} \quad (14)$$

2.4. Expression for shunt resistance

Similarly, expression for the shunt resistance R_{sh} can also be derived through differentiation. At the short circuit point, the derivative of current with respect to voltage satisfies (15) [9–13]. Therefore, taking the derivative of (2) at the short circuit point yields (16). Then the expression for the shunt resistance can be obtained from (16), given as (17). Furthermore, (17) can be simplified as (18) due to $R_s \ll R_{sh}$.

$$\frac{dI}{dV} \Big|_{I=I_{sc}} \approx -\frac{1}{R_{sh}} \quad (15)$$

$$\frac{1}{R_{sh}} \approx I_0 \exp(\frac{I_{sc} R_s}{nV_T}) (\frac{1 - \frac{1}{R_{sh}} R_s}{nV_T}) + \frac{1 - \frac{1}{R_{sh}} R_s}{R_{sh}} \quad (16)$$

$$R_{sh} = \frac{R_s}{2} + [\frac{R_s^2}{4} + \frac{R_s}{I_0 \exp(\frac{I_{sc} R_s}{nV_T}) \frac{1}{nV_T}}]^{1/2} \quad (17)$$

$$R_{sh} \approx [\frac{I_{sc}}{I_0 \exp(1)}]^{1/2} R_s \quad (18)$$

2.5. Constraint condition

The previous expressions are obtained under the assumption that the shunt resistance is relatively large. In order to satisfy this assumption, we added the constraint condition, given as (19). This condition specifies that the current passing through the shunt resistor at the open circuit point should be within one percent of the short circuit current.

$$(\frac{V_{oc}}{R_{sh}})/I_{sc} < 1\% \quad (19)$$

2.6. Expressions for all the parameters

Finally, for convenience, the calculation expressions for all equivalent circuit parameters, including the constraint condition, are summarized in (20).

$$\begin{cases} I_{ph} \approx I_{sc} \\ n \approx \frac{I_{sc}(V_{oc} - V_m)}{[I_m + I_{sc} \ln(\frac{I_{sc}}{I_{sc} - I_m})]V_T} \\ I_0 \approx I_{sc} / \exp(\frac{V_{oc}}{nV_T}) \\ R_s \approx (nV_T)/I_{sc} \\ R_{sh} \approx [\frac{I_{sc}}{I_0 \exp(1)}]^{1/2} R_s \\ (\frac{V_{oc}}{R_{sh}})/I_{sc} < 1\% \end{cases} \quad (20)$$

3. Results and discussion

3.1. Results under standard test conditions

To verify the performance of the proposed method, we applied our method to five photovoltaic (PV) modules and nearly 8000 solar cells. The PV modules are passivated emitter and rear contact (PERC) monocrystalline silicon solar cell modules, and are composed of 72 solar cells in series and 2 solar cells in parallel. The solar cells are traditional

back surface field (BSF) solar cells. They are all tested under standard test conditions (STC, 1000 W/m², AM 1.5, 25 °C). Table 2 lists the calculated results for the PV modules, as well as for an individual solar cell within each module. Fig. 2 shows the distribution histograms of four equivalent circuit parameters for the BSF solar cells. As shown in Table 2 and Fig. 2, the equivalent circuit parameters can be extracted correctly using our method.

3.2. Comparison with other methods

As the SM55 PV module has been extensively investigated [15,19–23], it is selected here for comparison. Under standard test conditions (STC), the four parameters of the SM55 module are: $V_{oc} = 21.7$ V, $I_{sc} = 3.45$ A, $V_m = 17.4$ V and $I_m = 3.15$ A. Table 3 presents the parameters of the SM55 module obtained by the proposed method and the methods in the references [15,19–23]. As shown in Table 3, the equivalent circuit parameters obtained by the proposed method are very close to those of the three numerical methods, with the exception of the shunt resistance. However, they differ from those of other analytical or combined methods, particularly for the reverse saturation current. Meanwhile, the root mean square error (RMSE) of the proposed method is larger than that of the three numerical methods but is comparable to that of the other analytical methods. This indicates that the accuracy of the proposed method, while inferior to that of the numerical methods, is on par with other analytical approaches.

Table 4 shows the simulated data of the SM55 module obtained by the proposed method. The measured data, which were extracted from the manufacturer's I-V curve using Engauge Digitizer software, are also presented in Table 4. Here, the absolute error (AE), mean absolute error (MAE), absolute percentage error (APE), mean absolute percentage error (MAPE) and root mean square error (RMSE) are defined by (21), (22), (23), (24) and (25) respectively. In order to reduce the impact of small currents, data points where $I < 0.1I_{sc}$ are omitted from APE and MAPE calculations. As shown in Table 4, the proposed method achieves a MAPE of about 0.7% and an RMSE of about 0.0167 A.

$$AE = |I_{sim} - I_{mea}| \quad (21)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n AE_i = \frac{1}{n} \sum_{i=1}^n |I_{i,sim} - I_{i,mea}| \quad (22)$$

$$APE = \frac{|I_{sim} - I_{mea}|}{I_{mea}} \quad (23)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n APE_i = \frac{1}{n} \sum_{i=1}^n \frac{|I_{i,sim} - I_{i,mea}|}{I_{i,mea}} \quad (24)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n |I_{i,sim} - I_{i,mea}|^2} \quad (25)$$

Table 2

Calculated equivalent circuit parameters of PV modules and an individual solar cell within each module.

No.	V_{oc} (V)	I_{sc} (A)	V_m (V)	I_m (A)	I_{ph} (A)	n	I_0 (nA)	R_s (Ω)	R_{sh} (Ω)	P (%)
1*	49.23	13.72	41.32	12.83	13.72	83.88	1.65	0.1571	8700.8	0.041
1**	0.684	6.860	0.574	6.415	6.860	1.165	0.82	0.00436	241.7	0.041
2*	49.46	13.79	41.48	12.90	13.79	84.50	1.76	0.1574	8458.3	0.042
2**	0.687	6.895	0.576	6.450	6.895	1.174	0.88	0.00437	235.0	0.042
3*	49.60	13.86	41.64	12.97	13.86	84.16	1.51	0.1560	9059.9	0.039
3**	0.689	6.930	0.578	6.485	6.930	1.169	0.76	0.00433	251.7	0.039
4*	49.76	13.93	41.80	13.04	13.93	84.04	1.37	0.1550	9498.3	0.038
4**	0.691	6.965	0.581	6.520	6.965	1.167	0.68	0.00431	263.8	0.038
5*	49.92	14.00	41.96	13.11	14.00	83.92	1.23	0.1540	9958.5	0.036
5**	0.693	7.000	0.583	6.555	7.000	1.166	0.62	0.00428	276.6	0.036

* PV modules

** Solar cells, $P = V_{oc}/R_{sh}/I_{sc}$

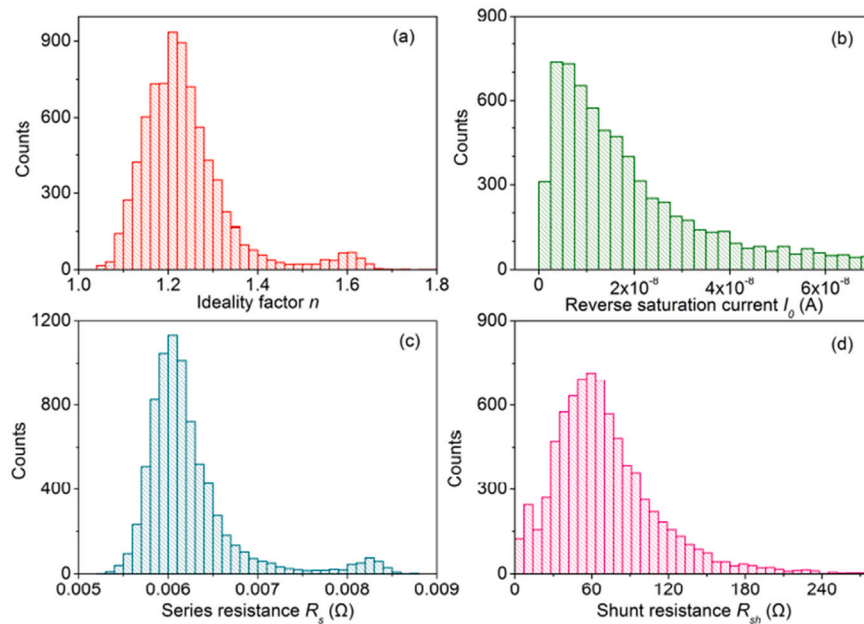


Fig. 2. Distribution histograms of the four equivalent circuit parameters for nearly 8000 solar cells.

Table 3

The parameters of the SM55 module obtained by the proposed method and the methods in the references.

Methods	Categories	I_{ph} (A)	n	I_0 (A)	R_s (Ω)	R_{sh} (Ω)	$RMSE$ (A)
The proposed	analytical	3.4500	49.8813	1.53E-7	0.3715	1070.8	0.01674
Reference [20]	numerical	3.4502	49.6115	1.36E-7	0.3390	451.2	0.00015
Reference [21]	numerical	3.4501	50.2470	1.71E-7	0.3291	483.9	0.00146
Reference [22]	numerical	3.4502	50.1079	1.67E-7	0.3301	480.8	0.00103
Reference [15]	analytical	3.4588	36.8075	4.06E-10	0.4954	334.1	0.02834
Reference [19]	combined*	3.4661	37.9008	7.49E-10	0.4752	295.4	0.01320
Reference [23]	combined*	3.4500	45.3600	2.75E-8	0.3600	233.5	0.03780

* combined analytical and numerical methods

Fig. 3 shows the comparisons of the measured data and simulated curves of the SM55 module under standard test conditions. As shown in Fig. 3, the simulation curves are in good agreement with the measured data points, and the absolute errors (AE) are relatively higher between 10 V and 21 V but lower near the short-circuit and open-circuit points.

Table 5 presents the parameters of the RTC France solar cell obtained by the proposed method and some metaheuristic algorithms in the references. As shown in Table 5, the equivalent circuit parameters obtained by the proposed method are less accurate than those of the metaheuristic algorithms; meanwhile, the RMSE of the proposed method is larger than that of the metaheuristic methods. Aside from the fact that the accuracy of the proposed method is inferior to that of metaheuristic algorithms, a more significant reason is that, in the RTC France solar cell case, the relative current error, i.e., $(V_{oc}/R_{sh})/I_{sc}$, reaches about 1.14%, which exceeds the range defined in Constraint (19).

3.3. Results under different irradiance and temperature conditions

The parameters of a photovoltaic module, named SM55-2, were extracted under different irradiance and temperature conditions. The short-circuit current of the SM55-2 module was found to be slightly different from that of the SM55 module. Table 6 shows the parameters of the SM55-2 module extracted under varying irradiance and temperature. Table 6 indicates that the proposed method can extract equivalent circuit parameters under different irradiance and temperature conditions.

Table 4

Simulated data of the SM55 module obtained by the proposed method.

V_{mea} (V)	I_{mea} (A)	I_{sim} (A)	AE (A)	APE (%)	AE ² (A ²)
0.0104	3.4497	3.4488	0.0009	0.0260	0.000001
2.4588	3.4412	3.4465	0.0053	0.1545	0.000028
4.9125	3.4355	3.4442	0.0087	0.2528	0.000075
7.4443	3.4298	3.4417	0.0119	0.3477	0.000142
9.9292	3.4184	3.4386	0.0202	0.5916	0.000409
13.0861	3.3917	3.4255	0.0338	0.9962	0.001142
15.5710	3.3287	3.3583	0.0296	0.8880	0.000874
17.0817	3.1932	3.1973	0.0041	0.1284	0.000017
17.3942	3.1485	3.1371	0.0114	0.3621	0.000130
17.6026	3.1066	3.0897	0.0169	0.5442	0.000286
17.8162	3.0563	3.0336	0.0227	0.7423	0.000515
18.0454	2.9934	2.9647	0.0287	0.9598	0.000825
18.2330	2.9208	2.9007	0.0201	0.6886	0.000404
18.6758	2.7252	2.7197	0.0055	0.2024	0.000030
19.0925	2.5045	2.5053	0.0008	0.0309	0.000001
19.4468	2.2936	2.2857	0.0079	0.3447	0.000063
19.7958	2.0129	2.0330	0.0201	1.0009	0.000406
20.2073	1.6735	1.6868	0.0133	0.7946	0.000177
20.6033	1.2894	1.3048	0.0154	1.1975	0.000238
20.8846	0.9835	1.0041	0.0206	2.0917	0.000423
21.2909	0.5156	0.5283	0.0127	2.4649	0.000162
21.5566	0.2013	0.1917	0.0096	*	0.000092
21.6972	0.0044	0.0057	0.0013	*	0.000002
			MAE	MAPE	RMSE
			0.0140	0.7052	0.0167 A

* $I < 0.1I_{sc}$

However, the RMSE values in the last column confirm that the highest accuracy is achieved under standard test conditions.

Fig. 4 shows the comparison of the measured and simulated curves of the SM55-2 module under different irradiance. Fig. 4 demonstrates that while the simulated curves agree well with the measured data across various irradiance conditions, the smallest absolute error for the current occurs under standard test conditions.

Fig. 5 shows the comparisons of the measured data and simulated curves of the SM55-2 module under different temperatures. Fig. 5 demonstrates that while the simulated curves agree well with the measured data across various temperature conditions, a greater

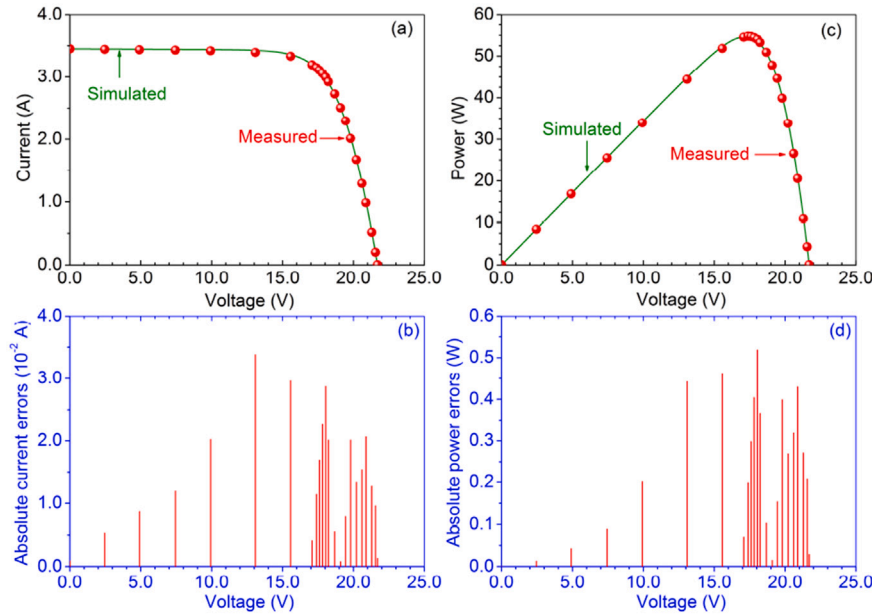


Fig. 3. Comparisons of the measured data and simulated curves of the SM55 module under STC: (a) I - V curve, (b) absolute current errors, (c) P - V curve, (d) absolute power errors.

Table 5

The parameters of the RTC France solar cell obtained by the proposed method and some metaheuristic algorithms in the references.

References	Algorithms	I_{ph} (A)	n	I_0 (A)	R_s (Ω)	R_{sh} (Ω)	$RMSE$ (A)
This paper	Proposed	0.7605	1.4484	1.54E-7	0.0489	65.99	5.56E-3
Reference [21]	ISCA	0.7608	1.4812	3.23E-7	0.0364	53.72	9.86E-4
Reference [22]	MPSO	0.7608	1.4753	3.11E-7	0.0365	52.89	7.73E-4
Reference [24]	GWOCs	0.7608	1.4808	3.22E-7	0.0364	53.63	9.86E-4
Reference [25]	MADE	0.7608	1.4812	3.23E-7	0.0364	53.72	9.86E-4
Reference [26]	CSBBO	0.7608	1.4812	3.23E-7	0.0364	53.72	9.86E-4

*ISCA - improved sine cosine algorithm; MPSO - mutated particle swarm optimization; GWOCs - grey wolf optimizer and cuckoo search; MADE - memetic adaptive differential evolution; CSBBO - cuckoo search biogeography-based optimization

Table 6

Parameters of the SM55-2 module extracted under varying irradiance and temperature.

Irradiance (W/m ²)	Temperature (°C)	I_{ph} (A)	n	I_0 (A)	R_s (Ω)	R_{sh} (Ω)	$RMSE$ (A)
1000	25	3.4178	47.3845	6.29E-08	0.3562	1592.8	0.0281
800	25	2.7326	45.1145	2.79E-08	0.4242	2546.8	0.0791
600	25	2.0660	45.2955	2.76E-08	0.5633	2955.0	0.0734
400	25	1.3623	40.9549	4.22E-09	0.7724	8420.1	0.0557
200	25	0.6956	40.1137	3.14E-09	1.4816	13,371.4	0.0328
1000	20	3.4171	49.0270	5.97E-08	0.3624	1663.0	0.0352
1000	30	3.4185	47.8062	1.29E-07	0.3653	1140.2	0.0396
1000	40	3.4325	47.6730	3.83E-07	0.3748	680.1	0.0719
1000	50	3.4465	46.5880	8.02E-07	0.3764	473.4	0.0776
1000	60	3.4605	47.0926	2.56E-06	0.3907	275.7	0.1083

deviation of the temperature from the standard test conditions results in a greater absolute error in the current. Note that the I - V parameters (including V_{oc} , I_{sc} , V_m , and I_m) of a photovoltaic module vary with irradiance and temperature, resulting in corresponding changes in the equivalent circuit parameters derived from these parameters.

3.4. Applicability and limitations

It is important to note that all methods based on simplified models or simplified equations, including the proposed method, should have specific scopes of applicability and cannot be applied to all scenarios. As indicated in the previous sections, the proposed method is based on the following four conditions: 1) single diode model, 2) extremely small I_0 , 3) relatively small R_s , and 4) relatively large R_{sh} . The fourth condition, relatively large R_{sh} , is the basis for simplifying the equations. However, not all solar cells satisfy this condition. Therefore, we introduce the constraint condition (19) to exclude non-conforming solar cells, which consequently narrows the applicability of the proposed method. Fig. 6 presents the relative current errors induced by the shunt resistance, as given by $(V_{oc}/R_{sh})/I_{sc}$, in the constituent cells of various PV modules, including RTC France, PWP201, SM55 and TP7F72M540. The data of the RTC France solar cell and the PWP201 PV module were obtained from Reference [24], and the TP7F72M540 PV module corresponds to the fifth entry in Table 2. As shown in Fig. 6, the SM55 and TP7F72M540 modules satisfy the constraint condition (19), while the RTC France and PWP201 modules do not.

Fig. 7 shows the percentage of relative current error in nearly 8000 solar cells, with an inset showing the distribution of errors less than 1%. Fig. 7 demonstrates that more than 95% of solar cells can satisfy the constraint condition (19). This indicates that, although the constraint limits the scope of the proposed method, it remains applicable to the vast majority of solar cells conforming to the single diode model.

Based on the above discussion, the limitations of the proposed method are summarized as follows. 1) The method is restricted to the single diode model; 2) It is not universally applicable to all solar cells, but only to more than 95% of them; 3) The applicable solar cells must be identified and verified according to Constraint (19).

3.5. Future research directions

The proposed method is based on single diode model. However, single diode model yields relatively large errors when analyzing solar cells with very high conversion efficiency, such as perovskite solar cells and tunnel oxide passivated contact (TOPCon) solar cells. Considering

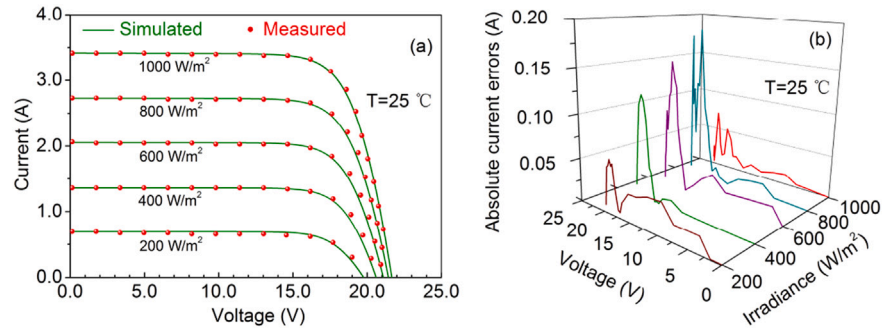


Fig. 4. Comparison of the measured data and simulated curves of the SM55-2 module under different irradiance: (a) I - V curve, (b) absolute current errors.

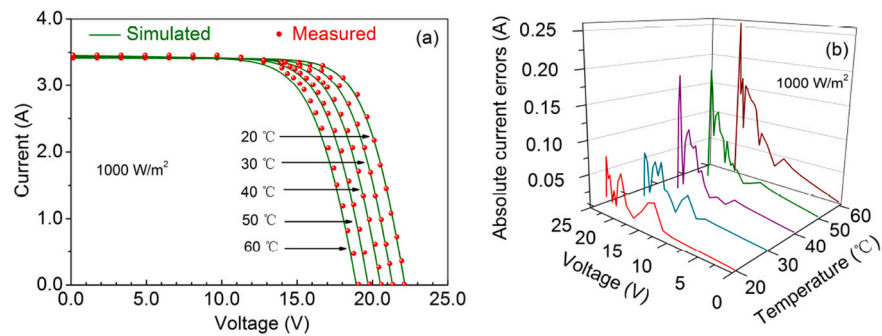


Fig. 5. Comparisons of the measured data and simulated curves of the SM55-2 module under different temperatures: (a) I - V curve, (b) absolute current errors.

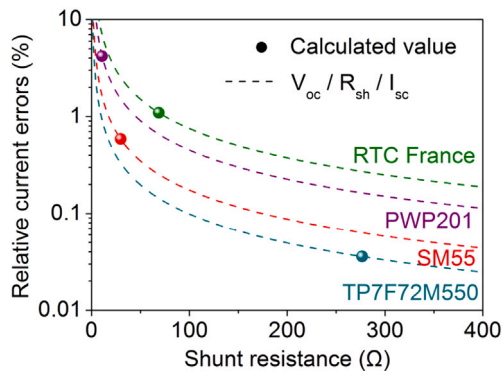


Fig. 6. Relative current error induced by the shunt resistance in the constituent cells of various PV modules.

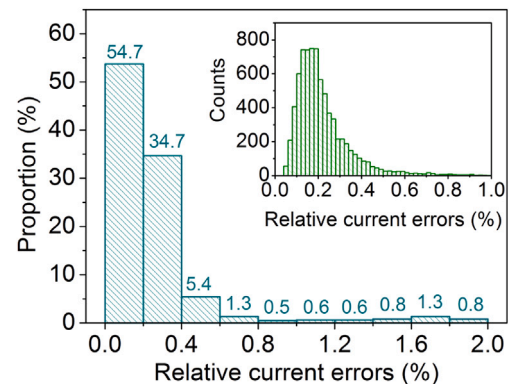


Fig. 7. Percentage of relative current error in nearly 8000 solar cells, the inset showing the distribution of errors less than 1%.

the simplicity of single diode model, a future research direction is to improve it for more accurate analysis of these two types of solar cells.

4. Conclusion

The equivalent circuit parameters of solar cells can be accurately estimated using simplified equations. This simplified equation-based method can overcome the drawback of existing analytical approaches that require the use of I - V curve slopes, while maintaining consistent estimation accuracy. Despite a slightly narrower applicable range, it remains applicable to over 95% of solar cells conforming to the single diode model.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Symbol meaning

To facilitate reference, all symbols used in the text and their meanings are listed in Table A.1.

Table A.1

Symbols and their meanings.

Symbols	Meanings	Symbols	Meanings
I	Current	R_s	Series resistance
I_{ph}	Photo generated current	R_{s0}	Reciprocal slope at OCP
I_0	Reverse saturation current	R_{sh}	Shunt resistance
I_{sc}	Short circuit current	R_{sh0}	Reciprocal slope at SCP
I_m	Maximum power point current	R_L	Load resistance
V	Voltage	n	Ideality factor
V_{oc}	Open circuit voltage	q	Elementary charge
V_m	Maximum power point voltage	k	Boltzmann constant
P	Power	T	Absolute temperature
P_m	Maximum power	V_T	Thermal Voltage

OCP: open circuit point; SCP: short circuit point.

Data availability

Data will be made available on request.

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